

## Site-Specific Photochemistry along a Protonated Peptide Scaffold

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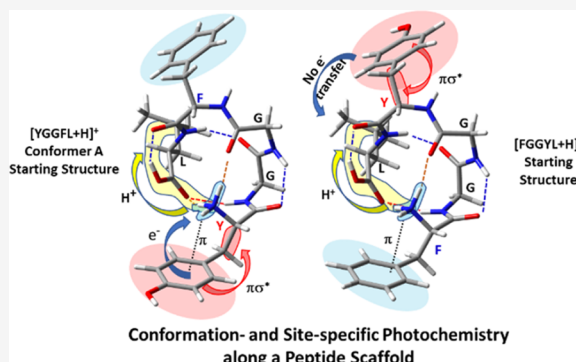


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**ABSTRACT:** We present a detailed study of the time-dependent photophysics and photochemistry of a known conformation of the two protonated pentapeptides Leu-enkephalin (Tyrosine-Glycine-Glycine-Phenylalanine-Leucine, YGGFL) and its chromophore-swapped analogue FGGYL, carried out under cryo-cooled conditions in the gas phase. Using ultraviolet–infrared (UV-IR) double resonance, we record excited state IR spectra as a function of time delay between UV and IR pulses. We identify unique Tyr OH stretch transitions due to the  $S_1$  state and the vibrationally excited triplet state(s) formed by intersystem crossing,  $T_n(v)$ . Photofragment mass spectra are recorded out of the  $S_1$  origin and following UV-IR double resonance. Several competing site-specific fragmentation pathways are discovered involving peptide backbone cleavage, Tyr side chain loss, and N-terminal  $NH_3$  loss mediated by electron transfer. In YGGFL, IR excitation in the  $S_1$  state promotes electron transfer (ET) from the aromatic ring to the N-terminal  $R-NH_3^+$  group leading to loss of neutral  $NH_3$ . This product channel is missing in FGGYL due to the larger distance for ET from Y(4) to  $NH_3^+$ . Selective loss of the Tyr side chain occurs out of an excited state process following UV excitation and is further enhanced by IR excitation in  $S_1$  and  $T_n(v)$  states of both YGGFL and FGGYL. Finally, IR excitation in the  $S_1$  or  $T_n(v)$  states fragments the peptide backbone exclusively at amide(4), producing the  $b_4$  cation. We postulate that this selective fragmentation results from intersystem crossing to produce vibrationally excited triplets with enough energy to launch the proton along a proton conduit present in the known starting structure.



## 1. INTRODUCTION

Much of our understanding of photodissociation derives from studies of small molecules in which the entire molecular framework (e.g., an extended  $\pi$  cloud of an aromatic molecule) is the electronic chromophore responsible for photoabsorption.<sup>1</sup> However, in larger molecules with significant conformational complexity, while electronic absorption often accesses excited states localized on the chromophore, the processes that follow need not stay localized. Indeed, both the absorption wavelength and the photochemical pathways themselves can be affected significantly by the site-specific chemical environment in which the chromophore resides.<sup>2</sup> Different conformations will locate the chromophore at different distances/orientations from specific reactive groups. The different local site environments will also modulate the energy differences between excited state surfaces and change the lifetime(s) of the initial and intermediate states involved in the photochemistry.

In the face of such complexity, there is a role for incisive experimental tests in which isomer and conformation-specific photochemistry can be explored under well-characterized conditions. Gas-phase ions offer one immediate advantage by removing solvent so that the intramolecular processes can be studied directly. Mass spectrometry has built into it a general

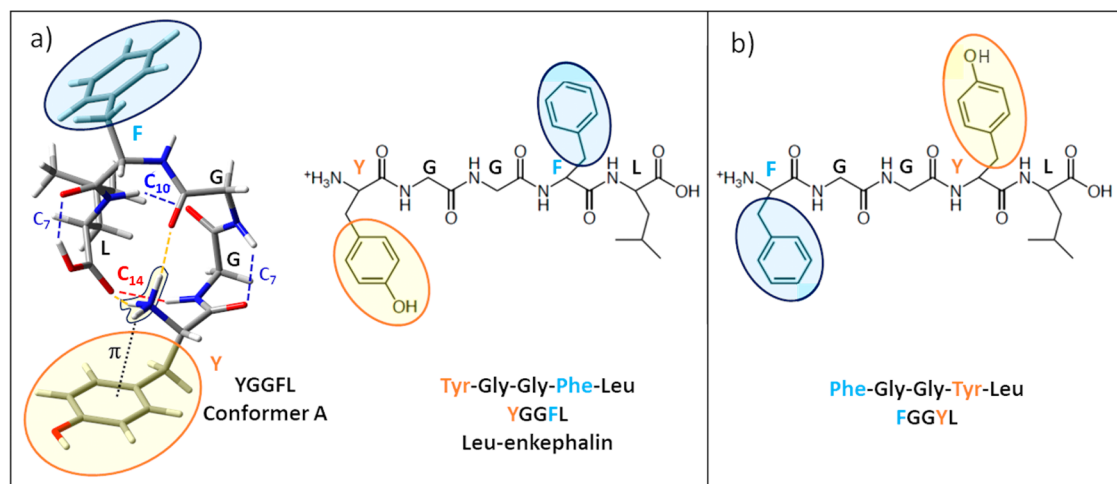
method for “purifying” reactants by mass selection and detecting photochemical products through changes in the mass-to-charge ratio following photoexcitation. Furthermore, if the ions are cooled to low temperatures prior to photochemical excitation, one can use the photofragmentation to record infrared (IR) and ultraviolet (UV) spectra of single conformations of the ion that serve as starting points for the photochemistry. The IR spectra are sensitive to the intramolecular H-bonding architecture of the conformation, enabling assignment to specific conformations.

Photochemistry of gas phase ions is often used as an alternative to collision-induced dissociation, providing complementary fragmentation behavior. Several groups have explored the UV photodissociation of aromatic-containing peptide ions at room temperature, often using  $MS^n$  methods to structurally characterize the fragment ions.<sup>3–9</sup> This includes an early study of protonated YGGFL at 280, 260, and 220 nm by

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**Figure 1.** Chemical structures of protonated Leu-enkephalin (YGGFL) (a) and its chromophore-switched isomer, protonated FGGYL (b). (a) Known three-dimensional structure of protonated YGGFL conformer A, adapted from ref<sup>16</sup>. C atoms are gray, O atoms red, N atoms blue, and H atoms white. The C<sub>n</sub> labels denote the number “n” of atoms separating the donor H from the acceptor O in an n-membered H-bonded cycle. The C<sub>10</sub> H-bond forms a  $\beta$ -turn in the peptide backbone.<sup>16</sup> Blue dashed lines represent XH...O=C H-bonds directed from N → C terminus, while red dashed lines point from C → N terminus.

Tabarin et al.<sup>5</sup> and a more detailed wavelength-dependent study seeking broad fragmentation coverage by Herburger et al.,<sup>9</sup> both conducted at room temperature. More recently, the groups of Fujii and Gregoire,<sup>2,10,11</sup> and Rizzo and Boyarkin<sup>12–14</sup> have used complementary methods to study the photochemistry of cryo-cooled tyrosine-containing peptide ions. The excellent recent review of Gregoire et al. distills much of this latter work into a coherent picture and points out where further studies are needed.<sup>2</sup>

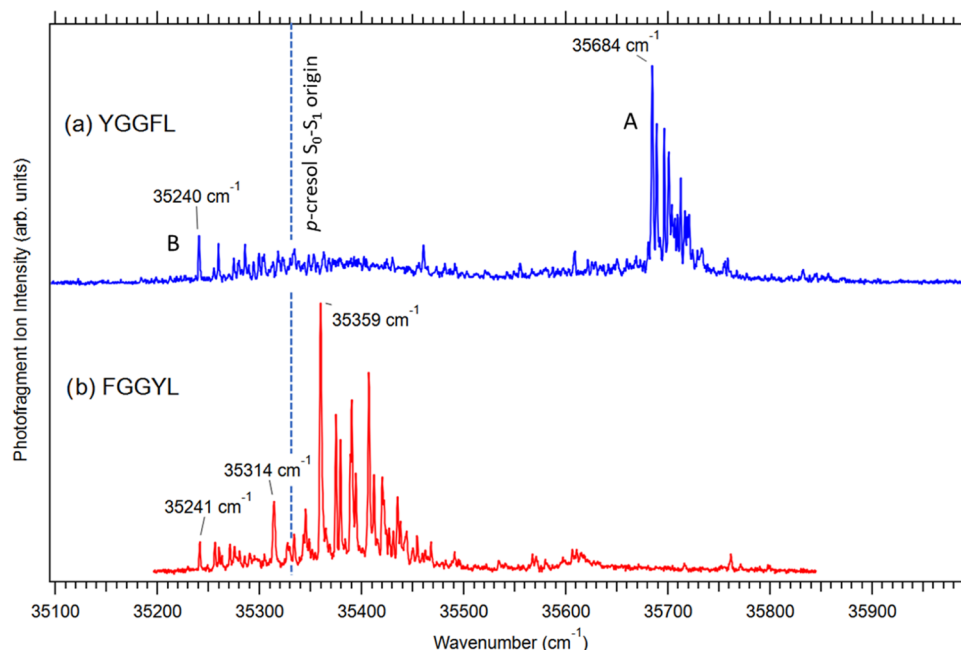
In this work, we use UV and IR-UV double-resonance methods to study the photochemistry of the model protonated peptide, Leu-enkephalin (Tyr-Gly-Gly-Phe-Leu, YGGFL) under cryo-cooled conditions in the gas phase, taking advantage of its well-characterized three-dimensional (3D) structure based on previous work from our group (Figure 1a).<sup>15,16</sup> The peptide backbone serves as a scaffold for studying the excited state dynamics initiated from the excitation of the tyrosine (Tyr or Y) chromophore in the precisely defined site environment of its single conformation structure, in which it engages with the nearby N-terminal protonation site through a  $\pi$  H-bond. We contrast those results with an analogous study of protonated FGGYL (Figure 1b), in which the phenylalanine and tyrosine chromophore positions are switched on an otherwise identical peptide backbone so that the Tyr chromophore takes up a site well removed from the N-terminus. In both pentapeptides, we focus our attention on the S<sub>1</sub> state in which the oscillator strength for the transition is localized on the Tyr chromophore. The S<sub>2</sub> state, in which excitation is localized primarily on the phenylalanine (F) chromophore, is more than 1300 cm<sup>−1</sup> to the blue of the S<sub>1</sub> origin,<sup>14</sup> and thus is not directly excited here.

The starting point for our work is to use UV photofragment mass spectra to explore the photochemical pathways following UV excitation of Y(1) site in YGGFL and Y(4) site in FGGYL. We then expand our studies by using UV-IR double-resonance spectroscopy to map out the IR spectrum of the excited state of each isomer as a function of delay between UV and IR lasers, thereby observing the effects of IR excitation in the first excited singlet (S<sub>1</sub>) and the vibrationally excited triplet states T<sub>n</sub>(v) formed by intersystem crossing. The use of excited state

IR spectra as a time-dependent structural probe is similar in spirit to the work of Rizzo and co-workers.<sup>12,14,16</sup> We discover unique spectroscopic signatures of the S<sub>1</sub> and triplet states of YGGFL and FGGYL via shifts in the OH stretch fundamental of the Tyr chromophore. At the electronic origins of both peptides, backbone fragmentation occurs indiscriminately following internal conversion to form hot ground state ions. Following IR absorption in the excited state(s), we observe several unique excited state processes, including (i) enhanced loss of the CH<sub>2</sub>–Ph–OH resonance-stabilized radical formed by cleavage of the Tyr C( $\alpha$ )–C( $\beta$ ) bond, (ii) products indicative of directed proton transfer (PT) along the peptide backbone, leading to breakage of a specific bond in the peptide backbone, and (iii) electron transfer (ET) from Tyr to NH<sub>3</sub><sup>+</sup>, leading to loss of neutral NH<sub>3</sub>. The results on FGGYL provide a marked contrast, eliminating the ET pathway, but retaining the bond-specific PT-driven backbone fragmentation. By combining the experimental results with calculations of the excited state dissociative pathways, we develop a scheme for the excited state dynamics that sheds new light on the site-specific processes involved.

## 2. EXPERIMENTAL METHODS

The experimental work was carried out in an apparatus described in detail elsewhere.<sup>15,17</sup> Nano-electrospray ionization (n-ESI) was used to bring the protonated peptides into the gas phase. The YGGFL sample was obtained commercially (95% purity) while the FGGYL sample was prepared at Fairfield University on a Protein Technologies PS3 peptide synthesizer using standard protocols employing Fmoc-Leu-Wang resin (Advanced Chemtech) on a 0.081 mmol scale. These samples were sprayed as a 50:50 H<sub>2</sub>O/CH<sub>3</sub>OH mixture at 100  $\mu$ M concentration. Mass selection of the parent was carried out following trapping in a room-temperature, linear quadrupole ion trap (QIT) using apex isolation, which also served as an annealing step for the ions before thermalization at room temperature. The mass-selected ions were transported to an octapole ion trap cooled to 5 K by a closed-cycle refrigerator. Helium pulsed into the cold trap serves as trapping gas and cools the ions into the conformational zero-point levels ( $T_{\text{vib}} \sim 10$  K). The output of a frequency-doubled neodymium-doped yttrium aluminum garnet (Nd:YAG) pumped dye laser (Pyrromethene 580/Rh6G mix) covering the 35,800–37,000 cm<sup>−1</sup> region was used to interrogate the cryo-cooled peptides in the region



**Figure 2.** UV photofragment spectra of (a) protonated Leu-enkephalin (YGGFL) and (b) its chromophore-switched isomer, FGGYL in the  $S_0$ – $S_1$  origin region of the Tyr chromophore. The wavenumber of the  $S_0$ – $S_1$  origin of *p*-cresol is shown for comparison, taken from ref31.<sup>31</sup>

of the Tyr  $S_0$ – $S_1$  origin. The ions were released from the trap to an analysis QIT (Q3), which operates in one of three modes. When recording a UV spectrum, resonant ejection is used to selectively remove parent ions from the trap (“parent knock-out”) prior to the collection of all remaining lower- $m/z$  UV photofragment ions independent of their mass. Alternatively, the mass spectrum of the photofragment ions can be recorded using mass-selective axial ejection (MSAE) at a fixed UV wavelength.<sup>18</sup> Finally, if the photofragment ion signal in a given mass channel is sufficiently large, mass-selective UV spectra can be recorded by operating the Q3 QIT in MSAE mode while gating on a specific product mass.

Ground state IR spectra are recorded by one of two methods. Conformer-specific IR spectra use IR-UV depletion spectroscopy, in which the UV wavelength was fixed on a single vibronic transition of the parent ion, producing a constant photofragment signal due to the single conformation responsible for this absorption. A tuned nanosecond Nd/YAG-pumped infrared parametric converter (Laser-Vision) is used as infrared source. Its output was spatially overlapped with the UV beam but temporally precedes the UV by  $\sim 100$  ns. When the IR wavelength is resonant with a vibrational transition of the monitored conformation, absorption removes a fraction of the single-conformer population in its zero-point level, leading to a dip or depletion in the UV photofragment ion signal.

Alternatively, IR-UV gain spectra provide ground state IR spectra that are not conformation-specific. As is particularly the case for YGGFL and FGGYL, the IR excited ions possess a broadened UV absorption that can show up as a gain signal if the additional internal energy enhances photofragmentation. This gain signal partially cancels the depletion signal when the UV laser is on-resonance, making it difficult to record high-quality IR-UV depletion spectra. However, when the UV laser is fixed off-resonance from any UV transitions of the cold ion, the broadened gain signal occurs against zero background and represents a composite IR spectrum containing abundance-weighted contributions from the conformations present. While the IR-UV gain spectra are no longer conformer-specific, when the IR dip and IR gain spectra are the same, it provides strong evidence that a single conformation of the ion is present. One can also remove the gain contributions from an IR-UV dip spectrum by recording IR-UV dip and IR-UV gain spectra under identical conditions and taking their difference. This is necessary when the

gain contributions are comparable to or greater than the dips we are seeking to observe.

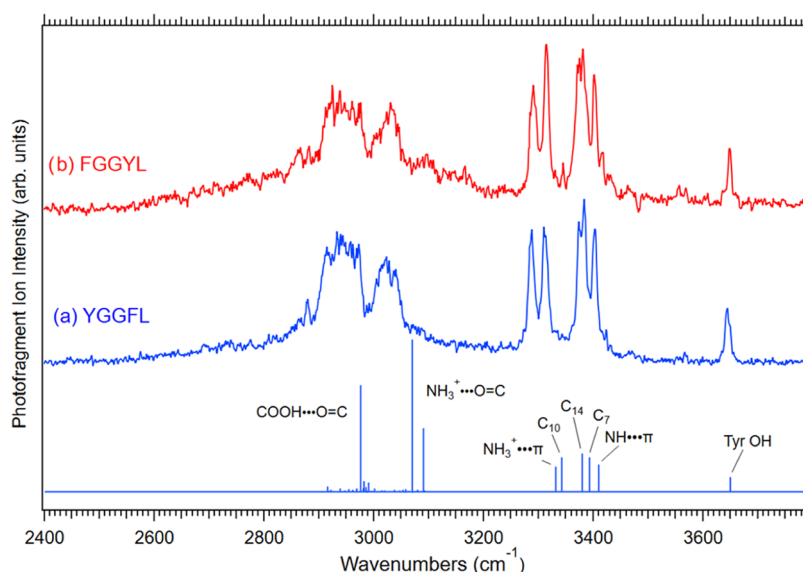
UV-IR double resonance was used to record IR spectra in the excited electronic state(s). A schematic diagram for the method is shown in Figure S1. This is accomplished by switching the order of excitation so that the IR laser is delayed relative to the UV laser. In the context of UV photofragment spectroscopy, the IR absorption is typically observed as a gain in photofragment signal due to enhanced fragmentation following IR excitation in the excited electronic state(s). If the UV laser wavelength is fixed on resonance with a UV transition due to a single conformation, these UV-IR gain spectra are single conformation IR spectra in the excited electronic state. Furthermore, by changing the delay between UV and IR lasers (6 ns pulse widths), one can map out excited state dynamics on the ns time scale appropriate for the pulsed lasers used here.

### 3. COMPUTATIONAL METHODS

Motivated initially by the observation of a second minor conformer of YGGFL, conformational searching was performed on protonated YGGFL using Grimme’s CREST program introduced by Pracht et al.<sup>19</sup> 781 structures were returned using the CREST search algorithm with the GFNn-xTB functional. These structures were sorted by energy and peptide backbone similarity. Select structures were subsequently optimized with the B3LYP density functional,<sup>20–23</sup> employing the D3(BJ)<sup>24</sup> dispersion correction and the 6-31+g(d) basis set,<sup>25–27</sup> as implemented in Gaussian 16.<sup>28</sup> Ground state ( $S_0$ ) state harmonic frequencies were computed with the same functional and basis set. All reported harmonic frequencies are scaled by 0.9723, which brings the  $S_0$  Tyr OH frequency of YGGFL conformer A into agreement with experiment.  $S_1$  and  $T_1$  vertical excitation energies were computed using the time-dependent density functional theory (TDDFT) formalism, employing the  $\omega$ B97X-D functional and aug-cc-pVDZ basis set. All reported excitation energies are scaled by 0.8593, which brings the  $S_0$ – $S_1$  origin of YGGFL A rotamer “a” into agreement with experiment. TDDFT calculations were performed in the Q-Chem 5.4 quantum chemistry package.<sup>29</sup>

### 4. RESULTS AND ANALYSIS

**4.1. UV Photofragment Spectra.** Figure 2 presents UV photofragment spectra of protonated YGGFL and FGGYL



**Figure 3.** Overview IR-UV gain spectra of (a) YGGFL conformer A recorded with UV  $-40\text{ cm}^{-1}$  to the red of the YGGFL A  $S_1$  origin ( $35,684\text{ cm}^{-1}$ ) and (b) FGGYL recorded with UV  $-40\text{ cm}^{-1}$  from the main FGGYL origin ( $35,359\text{ cm}^{-1}$ ). The stick diagram below (a) is the calculated scaled harmonic IR spectrum of the assigned structure of YGGFL A. The near-identical spectrum of FGGYL relative to YGGFL A proves that switching the positions of the Tyr and Phe chromophores leaves the H-bonding architecture of the ion unchanged.

over the  $35,100\text{--}36,000\text{ cm}^{-1}$  region where the  $S_0\text{--}S_1$  origin due to the Tyr chromophore in the main conformer of YGGFL is known to occur.<sup>16</sup> For comparison, the  $S_0\text{--}S_1$  origin of *para*-cresol (*p*-methyl phenol),<sup>30</sup> a useful proxy for the isolated, unshifted neutral UV chromophore of Tyr, is indicated in the figure by a dashed line. In the present study, we focus only on the Tyr chromophore rather than phenylalanine (F), whose electronic origin is shifted more than  $2000\text{ cm}^{-1}$  to the blue and is much weaker, making its transitions imbedded among high vibronic levels of the Tyr.

Previous studies by our group<sup>15,16</sup> identified and spectroscopically characterized the main conformer of  $[\text{YGGFL} + \text{H}]^+$ , labeled as conformer A in Figure 2a, with  $S_0\text{--}S_1$  origin at  $35,684\text{ cm}^{-1}$ . A dense set of vibronic transitions built off this origin is due in part to Franck–Condon activity in low-frequency vibrations. In addition, there is a probable contribution from conformations that share the same peptide backbone architecture but differ in side chain rotamer. Our infrared spectra in the hydride stretch region are sensitive to the hydrogen-bonding architecture of the ion, but insensitive to these side chain conformations.

In the course of the present work, we extended our UV scan below  $35,500\text{ cm}^{-1}$  based on the presence of small transitions in the UV spectrum of YGGFL to the red of conformer A's origin and found transitions due to a second minor conformer (B) with electronic origin at  $35,240\text{ cm}^{-1}$ , labeled B in Figure 2a. The transitions due to YGGFL(B) change their intensity relative to those of A depending on the electrospray and parent isolation conditions.

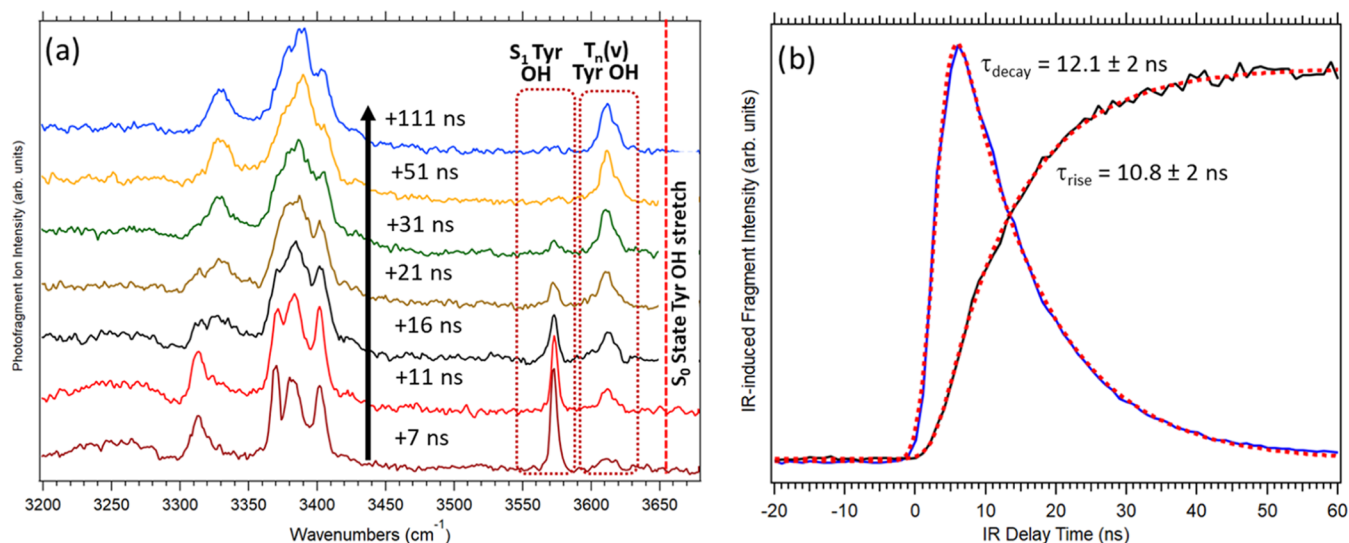
The UV photofragment spectrum of protonated FGGYL (Figure 2b) has its most intense electronic transition at  $35,359\text{ cm}^{-1}$ , shifted  $325\text{ cm}^{-1}$  to the red of the  $S_1$  origin of conformer A of YGGFL. In addition, there is a weaker set of transitions built off of an apparent electronic origin at  $35,314\text{ cm}^{-1}$ . As Figure S2 shows, this UV transition shares the same IR-UV dip spectrum as the main conformer at  $35,359\text{ cm}^{-1}$ . In what follows, we assign the electronic origin at  $35,314\text{ cm}^{-1}$  to the second OH rotamer of the Tyr side chain in FGGYL with the

peptide backbone retaining the same H-bonding architecture as the main conformer. The very weak transition at  $35,241\text{ cm}^{-1}$  has not been characterized but given its close proximity to the UV origin of conformer B of YGGFL ( $35,240\text{ cm}^{-1}$ ), the two may share a similar configuration that is quite different from the main conformers.

**4.2. IR-UV Dip and Gain Spectra.** The ground state IR spectrum of YGGFL A is shown in Figure 3a, recorded as an IR-UV gain spectrum, with UV laser  $40\text{ cm}^{-1}$  to the red of the  $S_1$  origin of YGGFL(A). The spectrum is identical to that in Burke et al., but in the course of the present work, we extended the spectrum from  $2400$  to  $3800\text{ cm}^{-1}$ . The corresponding IR-UV dip spectrum<sup>15</sup> is identical to the gain spectrum, indicating that at this UV wavelength, gain signals due to conformer A dominate. The stick diagram shown below the experimental spectrum in Figure 3a displays the calculated infrared intensities and scaled, harmonic vibrational frequencies for the previously assigned charge-stabilized  $\beta$ -turn structure of YGGFL shown in Figure 1a.<sup>15,16</sup> This structure is the primary peptide scaffold that we use as starting structure to explore the conformer-specific UV photochemistry of protonated YGGFL.

As a minor conformer, the IR-UV spectrum of YGGFL B was contaminated by significant photofragment gain signals due to conformer A both in dip (i.e., with UV fixed at the  $S_1$  origin of B,  $35,240\text{ cm}^{-1}$ ) and gain modes (with UV just off-resonance from the B origin). We have obtained a preliminary IR spectrum for conformer B as the difference signal between dip and gain spectra (Figure S3). However, the limited signal-to-noise and large interference made it difficult to assign this conformer with certainty. As a result, we concentrate in what follows on comparing the spectra and photochemistry of YGGFL conformer A with the corresponding conformer of FGGYL.

The IR-UV gain spectrum of FGGYL is shown in Figure 3b, where it is compared with that of YGGFL A. The spectrum was recorded with UV fixed just off-resonance ( $-40\text{ cm}^{-1}$ ) from the most intense transition in the UV photofragment spectrum ( $35,359\text{ cm}^{-1}$ ). In Figure S4, we compare the IR gain

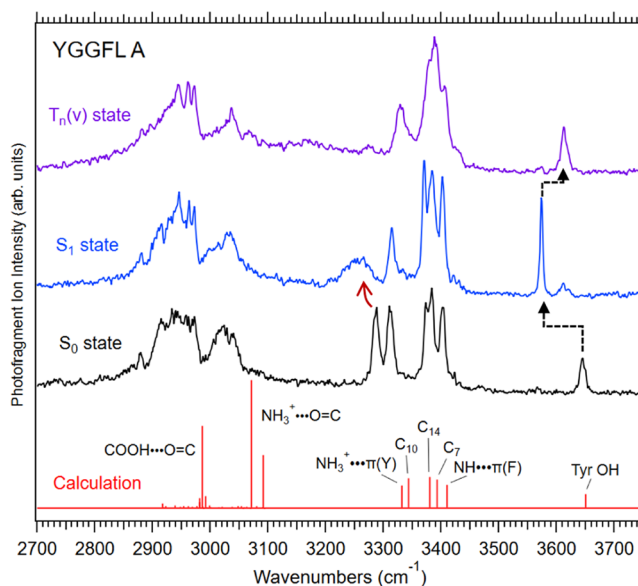


**Figure 4.** (a) Series of UV-IR gain spectra of  $[YGGFL + H]^+$  recorded with UV laser fixed on the  $S_0$ – $S_1$  origin of YGGFL(A) at  $35,684\text{ cm}^{-1}$ . The time labels indicate the time delay between IR and UV pulses,  $\Delta t = t_{\text{IR}} - t_{\text{UV}}$ . (b) Scans of the time delay between IR and UV pulses while monitoring the IR-enhanced total photofragment signal with IR at the peak of the  $S_1$  state Tyr OH stretch fundamental ( $3576\text{ cm}^{-1}$ , blue) and at the peak of the band assigned to the long-lived triplet state ( $3615\text{ cm}^{-1}$ , black). Fits to the data are shown as dotted lines with the indicated time constants. Details of the fitting procedure are in the [Supporting Information](#).

spectrum of FGYYL with the IR-UV dip spectrum with UV fixed at  $35,359\text{ cm}^{-1}$ , which are identical to one another. This demonstrates that all conformers that contribute to the UV gain signal carry the same IR spectrum and thus are due to the same H-bonding architecture. Note that the gain spectrum is nearly identical to that of YGGFL A, proving that swapping the positions of the Tyr and Phe chromophores did not disturb the H-bonding architecture, creating a circumstance in which we can explore the site-specific photochemistry of YGGFL A with FGYYL starting from the identical peptide conformation, shown in [Figure 1a](#).

**4.3. UV-IR Spectra and Timing Scans.** [Figure 4a](#) presents a series of UV-IR gain spectra involving IR excitation out of the  $S_1$  zero-point level of conformer A of YGGFL with IR tuned over the  $3200$ – $3700\text{ cm}^{-1}$  region that includes the amide NH stretch and Tyr OH stretch regions. These scans provide a means of recording excited state IR spectra of YGGFL A through the enhancement in the photofragment signal when the IR laser is resonant with an IR transition of the excited state of YGGFL A. The wavenumber position of the ground state Tyr OH stretch fundamental ( $3649\text{ cm}^{-1}$ , [Figure 5](#),  $S_0$ ) is indicated in [Figure 4a](#) by the red dashed line.

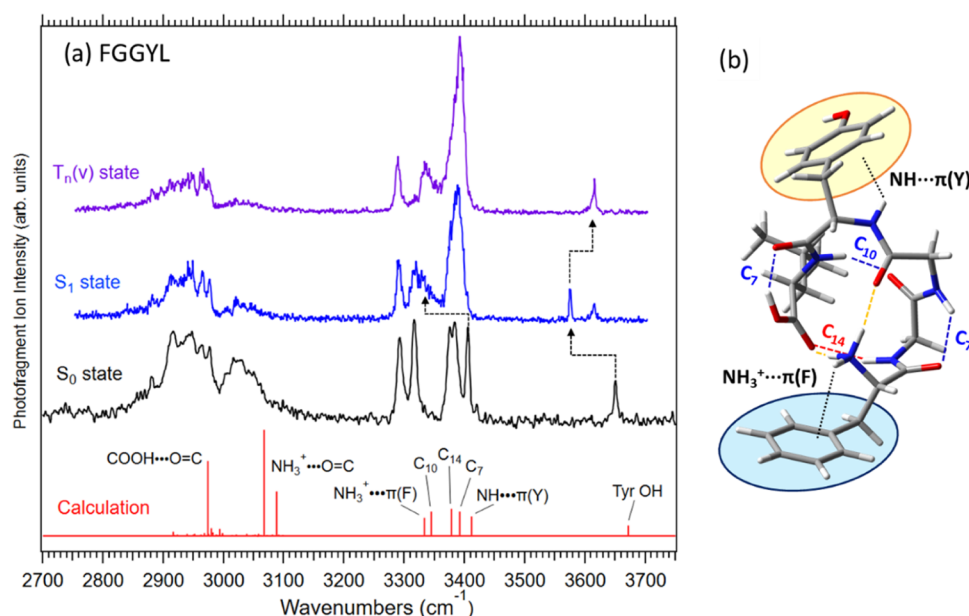
We observed that the disappearance of the ground state absorption is coincident with the appearance of a new IR transition shifted down to  $3576\text{ cm}^{-1}$ , a shift of  $-73\text{ cm}^{-1}$  from the ground state Tyr OH stretch fundamental ( $3649\text{ cm}^{-1}$ ). We recorded a series of UV-IR spectra as a function of delay time, noticing the fast drop-off in the intensity of the transition at  $3576\text{ cm}^{-1}$  and the appearance of a second transition at  $3615\text{ cm}^{-1}$  that grows in on the time scale that the band at  $3576\text{ cm}^{-1}$  decays. Consistent with a previous study by Scutelnic et al.,<sup>14</sup> the band at  $3576\text{ cm}^{-1}$  is assigned to the  $S_1$  state of YGGFL(A). We have also recorded time delay scans while monitoring the  $S_1$  OH stretch and the  $T_1(v)$  OH stretch transitions, monitoring the magnitude of these ion gain signals as a function of delay between the UV excitation and IR probe pulse, which are shown in [Figure 4b](#). Given the 6 ns pulse durations of our IR and UV laser pulses, we formed a cross-



**Figure 5.** Overview IR-UV gain (black), UV-IR gain with time delay  $\Delta t = +7\text{ ns}$  (blue), and UV-IR gain spectrum at a time delay of  $+50\text{ ns}$  (purple) which probe the  $S_0$ ,  $S_1$ , and  $T_n(v)$  states, respectively, of  $[YGGFL + H]^+$  conformer A. The assignment labels on the calculated stick diagram of the  $S_0$  state spectrum are taken from ref<sup>17,16</sup>. See [Figure 1](#) to interpret the “ $C_n$ ” labels in terms of amide  $\text{NH}\cdots\text{O}=\text{C}$  H-bonds, and the text for further discussion.

correlation function of the UV and IR laser pulses (measured with fast photodiodes) and convolved this function with a single exponential decay for the  $S_1$  state signal and single exponential rise for the triplet state signal. In YGGFL(A) the best fit to the decay of the band at  $3576\text{ cm}^{-1}$  is for  $\tau_{\text{decay}} = 12.1 \pm 2\text{ ns}$ , a value similar to that in other Tyr-containing peptides.<sup>14,32</sup>

The rise time of the band at  $3615\text{ cm}^{-1}$  is  $\tau_{\text{rise}} = 10.8 \pm 2\text{ ns}$ . This rise/decay is dictated by the sum of the rate constants out of the  $S_1$ -state origin, which includes internal conversion and



**Figure 6.** (a) Overview IR-UV dip spectrum (black, flipped to appear as a gain), UV-IR gain spectrum of FGGYL with time delay of +7 ns (blue), and UV-IR gain spectrum at a time delay of +50 ns (purple) which probe the  $S_0$ ,  $S_1$ , and  $T_n(v)$  states, respectively, of the main conformer of  $[FGGYL + H]^+$  shown to the right. All spectra were recorded with UV wavelength fixed on the  $S_0$ – $S_1$  origin of FGGYL. The arrows connect transitions that shift with electronic excitation, most notably the Tyr OH stretch and the NH– $\pi$ (Y) stretch. The “ $C_n$ ” labels indicate the specific H-bonded cycle of “ $n$ ” atoms closed by the amide  $NH\cdots O=C$  H-bonds shown in the figure to the right. The stick diagram below the experimental scans shows the calculated scaled harmonic vibrational frequencies and IR intensities in the  $S_0$  state at the B3LYP-D3(BJ)/6-31+G(d) level of theory. (b) Assigned structure for FGGYL, identical to that for YGGFL A but with the positions of the two electronic chromophores, Phe(F) and Tyr(Y) switched.

direct fragmentation of the Tyr side chain (Section 4.4). We associate the  $3615\text{ cm}^{-1}$  transition with the OH stretch fundamental in the triplet state,  $T_n(v)$ . Based on the relative intensities of fragments in the photofragment mass spectra, we estimate a rate constant for intersystem crossing of  $k_{ISC} \sim 2$  to  $5 \times 10^7\text{ s}^{-1}$ . Furthermore, the transition at  $3615\text{ cm}^{-1}$  decays away on a much longer time scale (Figure S5), with a best-fit time constant of  $20 \pm 2\text{ }\mu\text{s}$ , a reasonable value for a second intersystem crossing of  $T_n(v)$  back to the singlet ground state,  $S_0$ . We therefore tentatively assign the transition at  $3615\text{ cm}^{-1}$  to the triplet state(s)  $T_n(v)$  of Tyr formed by intersystem crossing (ISC) from the  $S_1$  state. The unusual breadth of the transition ( $\Delta E_{\text{fwhm}} \sim 20\text{ cm}^{-1}$ ) is thought to arise from the vibrational energy of the triplet states accessed by intersystem crossing (ISC), equal to the energy difference between the adiabatic origins of the  $S_1$  and  $T_1$  states,  $E_{\text{vib}} = \Delta E(S_1 - T_1)$ .

Having established conditions under which we can record IR spectra due to  $S_1$  ( $\Delta t = +7\text{ ns}$ ) and  $T_n(v)$  ( $\Delta t = +50\text{ ns}$ ) excited states of the Tyr chromophore in YGGFL(A). The  $S_1$  state IR spectrum is shown as the blue trace in Figure 5, while the triplet state spectrum is shown above it as a purple trace. For most vibrational bands, the spectra are similar to one another and to the  $S_0$  state spectrum (black trace). This indicates that on the time scale that we record the excited state IR spectra, the H-bonding architecture of YGGFL A is largely retained, with absorptions due to the  $COOH\cdots O=C$  and the  $C_{10}$  NH stretch that forms the  $\beta$ -turn, intact. Nevertheless, the transition assigned to the  $NH_3^+\cdots\pi$  H-bond in the  $S_1$ -state spectrum does show a broadening and shift to a lower frequency ( $3255\text{ cm}^{-1}$ ) relative to the corresponding transition in  $S_0$  at  $3287\text{ cm}^{-1}$ . It seems logical that this band should be particularly sensitive to electronic excitation of the Tyr ring, with its breadth arising from the changed  $NH_3^+\cdots\pi(Y)$  distance

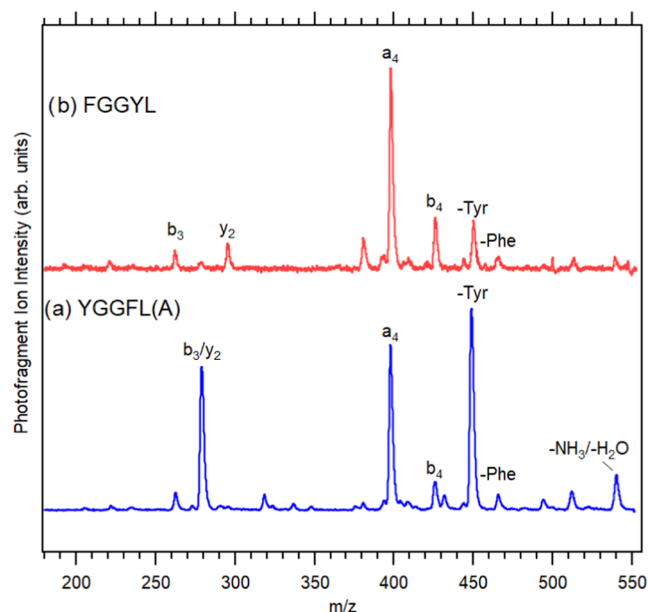
spreading the NH stretch intensity over many low-frequency combination bands. At longer delay times of the  $T_n(v)$  spectrum, all of the bands assigned to the  $NH_3^+$  group are replaced by a broad, nearly featureless absorption over the  $3100$ – $3300\text{ cm}^{-1}$  region in the  $T_n(v)$  spectrum, indicating that the entire binding pocket for the  $NH_3^+$  group has been affected by the combination of electronic excitation and excess vibrational energy associated with the intersystem crossing. The amide NH stretch fundamentals show shifts of a few  $\text{cm}^{-1}$  and broaden in the triplet state spectrum.

We carried out a similar exploration of excited state processes in FGGYL. As in YGGFL(A), the corresponding conformation of protonated FGGYL shows a shifted  $S_1$ -state OH stretch at  $3578\text{ cm}^{-1}$  and a  $T_n(v)$  state OH stretch at  $3615\text{ cm}^{-1}$ , with similar best-fit decay ( $12.5 \pm 3\text{ ns}$ ) and rise-times  $8.3 \pm 3\text{ ns}$  (Figure S6), to those in YGGFL A.

The corresponding state-specific IR spectra for FGGYL are shown in Figure 6a. The  $S_1$  and  $T_n(v)$  state spectra were recorded with UV fixed on the  $S_1$  origin at  $35,359\text{ cm}^{-1}$ . The  $S_0$  spectrum (black) is reproduced below them for comparison, with the calculated spectrum for the assigned structure below it. The Tyr OH stretch fundamental shifts in frequency with electronic state just as it does in YGGFL(A). Since the  $S_1$  state spectrum was recorded at a timing that maximized the  $S_1$ -state OH stretch, with the IR laser delayed by +7 ns from the UV laser, the  $T_n(v)$  OH stretch is beginning to appear in the  $S_1$  state spectrum, but then becomes the only OH stretch transition by a delay of +50 ns in which the  $T_n(v)$  state spectrum was recorded. Overall, the  $S_1$  and  $T_n(v)$  spectra of FGGYL are quite similar to one another, with fewer changes to the overall appearance than in YGGFL A. The most notable change between  $S_0$  and  $S_1/T_n(v)$  spectra is in the frequency of the NH– $\pi$ (Y) stretch, which appears at  $3405\text{ cm}^{-1}$  in the  $S_0$

state. We tentatively assign the shifted  $S_1$ -state transition to the shoulder at  $3331\text{ cm}^{-1}$ .

**4.4. Photofragment Mass Spectra.** **4.4.1. Photochemistry at the  $S_1$  Origin.** With the assignments for the starting conformational structures of protonated YGGFL(A) and FGGYL in hand, we recorded photofragment mass spectra following UV excitation at their respective  $S_1$  origins. The results are shown in Figure 7. The relative intensities of mass



**Figure 7.** UV photofragment mass spectra with UV fixed on the (a)  $S_0$ – $S_1$  origin of YGGFL conformer A ( $35,684\text{ cm}^{-1}$ ) and (b) the corresponding  $S_0$ – $S_1$  origin of the main conformer of FGGYL ( $35,359\text{ cm}^{-1}$ ). Both mass spectra have a background scan subtracted from them without UV present. Important peaks in the mass spectra are labeled, with  $-X$  indicating the loss of the neutral  $X$  from the  $MH^+$  parent ion ( $m/z$  556 Da).  $-Tyr$  is loss of the Tyr side chain, ( $\dot{C}H_2$ –Ph–OH,  $MH^+$ –107). The  $a/b/y$  designations are peptide backbone fragment peaks. See the main text for further discussion.

peaks in the photofragment mass spectra vary to some degree with settings used for bringing ions into and out of the cold trap. Therefore, in what follows, we refer to trends in photofragment product distributions rather than giving explicit percentage product yields. To record the mass spectra in Figure 7, we used a scan function for the analysis quadrupole ion trap (Q3) that scanned from low mass to within a few Da of the parent ion ( $m/z$  556 Da). We report the difference mass spectrum with and without the UV laser present.

The mass spectrum due to YGGFL A shows large photofragment ion signals associated with the loss of the Tyr side chain ( $MH^+$ –107 = 449 Da) and a peptide backbone fragment ( $a_4^+$ ) commonly present in the collision-induced or surface-induced dissociation spectra of protonated YGGFL.<sup>33</sup> The labeling scheme and fragment identification are shown in Figure S7. Previous UV photofragmentation studies of other Tyr-containing protonated peptide ions have identified the Tyr side chain loss as characteristic of an excited state process, as this fragment is not present in CID studies.<sup>2,12,32</sup> By contrast, the peptide backbone fragments ( $a/b/y$ ) are associated with fragmentation following internal conversion to the ground state to form a hot ground state ion with  $35,684\text{ cm}^{-1}$  of initial internal energy. Due to the swap in Phe(F) and Tyr(F)

chromophores, the  $b_3$  and  $y_2$  masses of the two isomers overlap in YGGFL but are well separated in FGGYL. Collisions with cold helium in the trap will remove this energy in competition with fragmentation. The ratio of  $a_4/b_4$  fragments is known to vary with the internal energy of the YGGFL parent ion, with higher  $a_4$  signal reflecting higher temperature.<sup>34</sup> The dominance of  $a_4$  over  $b_4$  in all three mass spectra is consistent with the highly non-Boltzmann internal energy of the ions following internal conversion. The ratio of intensities of the  $-Tyr$  and  $a_4$  fragment peaks changes significantly between YGGFL A and FGGYL; however, since  $a_4$ ,  $b_3$ , and  $y_2$  all arise from internal conversion to the ground state, the sum of the intensities of these three peaks relative to  $-Tyr$  is more nearly consistent between them. We surmise that at the electronic origins of both YGGFL A and FGGYL, peptide backbone fragmentation following internal conversion is the dominant process, followed by loss of the Tyr side chain.

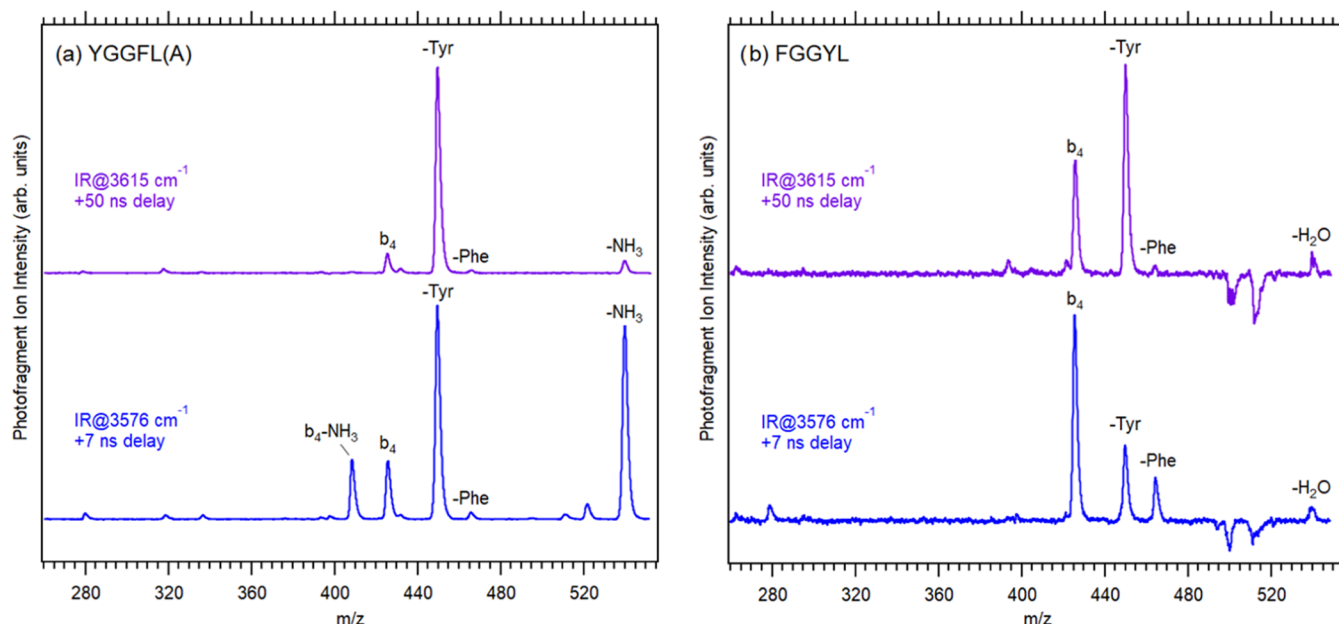
**4.4.2. Photochemistry Following IR Excitation in the Excited State(s).** The series of photofragment mass spectra shown in Figure 8 were taken following UV-IR double-resonance excitation, and are difference spectra [(UV-IR) – (UV-only)] with UV wavelength fixed on the  $S_0$ – $S_1$  origins of the indicated peptide and IR excitation of either the  $S_1$  OH stretch ( $3576\text{ cm}^{-1}$ ) or  $T_n(v)$  OH stretch ( $3615\text{ cm}^{-1}$ ). In both YGGFL(A) and FGGYL, the blue traces were recorded at a delay that maximizes the  $S_1$  OH stretch intensity (+7 ns) and are therefore characteristic of fragmentation following infrared excitation in the  $S_1$  state. The purple traces were recorded under triplet state conditions, with a +50 ns delay between IR and UV.

The IR-induced photofragment mass spectra in Figure 8 are quite simple compared to those at the  $S_1$  origin (Figure 7). In all cases, the Tyr loss channel is significantly enhanced by IR excitation, independent of which peptide is excited and whether the ion is in the  $S_1$  or  $T_n(v)$  states. We surmise on this basis that the excited state responsible for loss of the Tyr side chain is energetically accessible and competes effectively with other fragmentation pathways under all of these circumstances.

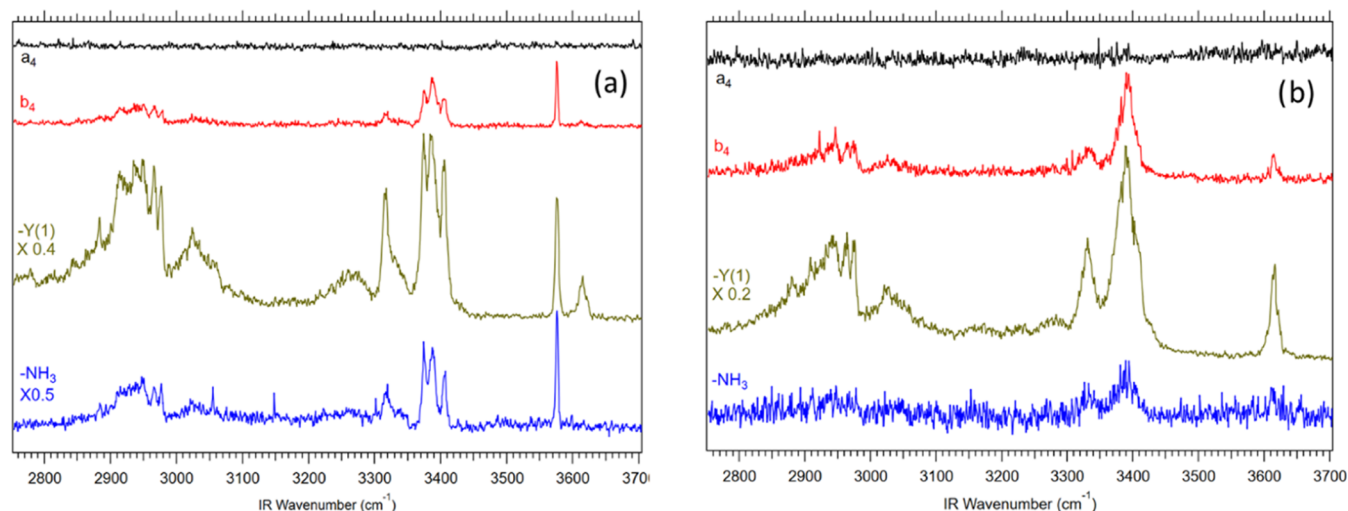
The results for YGGFL(A) in Figure 8a are particularly notable in another regard. Following IR excitation in the  $S_1$  state, a large photofragment signal appears at  $[MH^+ - 17]$  associated with loss of neutral  $NH_3$ . We have looked carefully at high mass resolution to confirm that the IR enhancement is due exclusively to  $NH_3$  loss rather than  $H_2O$  loss (Figure S8). The intensity of this fragment channel in YGGFL A is comparable in size with the Tyr side chain loss. Interestingly, the mass spectrum recorded with +50 ns delay shows no  $NH_3$  loss, but only loss of the Tyr side chain.

By contrast, in FGGYL, the  $NH_3$  loss channel is missing entirely under both  $S_1$  and  $T_n(v)$  conditions. We conclude based on these mass spectra that, following IR excitation in the  $S_1$  state, YGGFL A reacts selectively and efficiently via an electron transfer (ET) reaction to the N-terminal  $R-NH_3^+$  group, forming neutral  $NH_3 + R^{\bullet+}$ , while FGGYL does not.

Finally, the photofragment mass spectra in Figure 8 are also highly selective in their backbone fragmentation. The  $a_4$ ,  $b_3$ , and  $y_2$  fragments ascribed to internal conversion are notably absent from any of the IR-enhanced UV-IR mass spectra, even though they are among the dominant photofragments at the  $S_1$  origin (Figure 7). Instead, in FGGYL (Figure 8b) only the  $b_4$  fragment is observed, associated with breaking the amide N–C bond joining Tyr(Y) and leucine (L), while in YGGFL (A) the



**Figure 8.** UV-IR double-resonance photofragment mass spectra following UV excitation of (a) conformer A of YGGFL and (b) the main conformer of FGGYL. The UV laser was fixed on the  $S_0$ – $S_1$  origin transitions of the two isomers. The mass spectra are the difference in ion signal between UV-IR and UV-only mass spectra, thereby highlighting those peaks that are enhanced by IR excitation in the excited state. Mass spectra recorded under  $S_1$ -state conditions (+7 ns delay of IR relative to UV, IR at  $3576\text{ cm}^{-1}$ , the peak of the  $S_1$  state OH stretch) are shown in blue, while mass spectra recorded under  $T_n(v)$  conditions (+50 ns delay, IR at  $3615\text{ cm}^{-1}$ , the peak of the  $T_n(v)$  OH stretch) are shown in purple. See text for further discussion.



**Figure 9.** UV-IR spectra of YGGFL A taken with the IR delayed from UV by (a) +7 ns and (b) +10  $\mu\text{s}$ , while simultaneously monitoring the indicated masses in the photofragment mass spectrum as the IR was scanned. The indicated product ions are representative of specific reaction pathways, including internal conversion ( $a_4$ ), proton transfer ( $b_4$ ), Tyr side chain loss ( $-Y(1)$ ), and electron transfer ( $-\text{NH}_3$ ).

$b_4^+-\text{NH}_3$  ion is also produced. This is a clear indication that the  $b_4^+$  and  $b_4^+-\text{NH}_3$  ions are not produced by internal conversion to the ground state but must arise from an electronically excited state. The fact that this specific backbone fragmentation is observed exclusively is highly unusual and deserves further consideration in Section 5.

**4.5. Mass-Resolved UV-IR Spectra.** The UV-IR spectra in Figures 5 and 6 measured the total photofragment ion signal, independent of mass, as the IR laser was scanned. In order to test whether there are differences in the way individual photofragments respond to IR excitation, we recorded a series of mass-resolved UV-IR spectra while simultaneously monitoring the integrated photofragment ion signals in the  $a_4$ ,  $b_4$ ,  $-$

$Y(1)$ , and  $-\text{NH}_3$  mass channels that are the major product channels. Figure 9a presents a series of mass-resolved UV-IR spectra out of the  $S_1$ -state of YGGFL(A), while Figure 9b has a delay between UV and IR excitation of 10  $\mu\text{s}$ , and therefore probes the long-lived intermediate state that we have assigned to the triplet state,  $T_n(v)$ . The analogous mass-resolved UV-IR spectra of FGGYL are shown in Figure S9.

Several deductions can be drawn from these spectra. First, since loss of the Tyr side chain is the dominant product channel in YGGFL A, the  $S_1$ -state UV-IR scan monitoring all product ions (Figure 5, blue trace) looks much like the mass-resolved IR spectrum monitoring Tyr loss ( $-Y(1)$ , Figure 9a). Second, as anticipated from the IR-enhanced mass spectra

(Figure 8), loss of neutral  $\text{NH}_3$  following electron transfer is greatly enhanced by IR excitation out of the  $S_1$  zero-point level, with its  $S_1$  state Tyr OH stretch transition nearly as large as that in the  $-\text{Y}(1)$  mass channel. Third, the relative intensities of UV-IR transitions in the  $\text{NH}_3$  loss channel are quite different than those in the Tyr loss channel. In particular, the  $\text{NH}_3$  loss channel has significantly smaller IR enhancements on IR transitions below  $3350\text{ cm}^{-1}$ , suggesting that transitions above this wavenumber have overcome an energetic threshold to accessing the charge transfer (CT) state associated with electron transfer to form neutral  $\text{NH}_3$ . Below this threshold, we postulate that electron transfer may be occurring by tunneling to this CT state. Fourth, the  $a_4$  mass channel that dominates photofragmentation at the  $S_1$  origin shows no IR enhancement in either  $S_1$  or  $T_n(v)$  conditions. We have checked this also to hold for the  $b_3$  and  $y_2$  fragment ions, confirming that all of these fragments are formed following internal conversion to the ground state. Fifth, mass-resolved UV-IR spectra for YGGFL(A) with IR excitation occurring  $10\text{ }\mu\text{s}$  after the UV pump pulse (Figure 9b) have the same two dominant products,  $-\text{Y}(1)$  and  $b_4$ , with similar relative intensities to those recorded when the  $T_n(v)$  OH stretch first appears ( $\tau = +50\text{ ns}$ ). Notably, the UV-IR spectrum of the  $b_4$  fragment mimics that of  $-\text{Y}(1)$ , including possessing a clear  $T_n(v)$  OH stretch transition at  $3615\text{ cm}^{-1}$ . This argues for these two products both being formed from the same intermediate state and that the OH stretch itself is characteristic of that common intermediate state. Finally, in FGGYL (Figure S9), loss of the Tyr side chain remains the dominant photofragment, despite the shift of the Tyr side chain from the (1) to (4) position on the peptide backbone.

**4.6. Excited State Calculations.** The assigned structures for YGGFL A and FGGYL are based on comparison of the ground state IR spectra with scaled, harmonic vibrational frequencies and IR intensities at the  $S_0$  optimized geometry. As a result, we are not able to distinguish between the two OH rotamers in which the Tyr OH, which lies in the plane of its aromatic ring, can point in either direction with respect to the peptide backbone. The UV photofragment spectra in Figure 2 show evidence for the presence of the two OH rotamers, especially in FGGYL where a second  $S_0-S_1$  electronic origin is shifted  $-46\text{ cm}^{-1}$  from the main origin at  $35,359\text{ cm}^{-1}$ .

In Table 1, we report scaled, vertical TD-DFT calculations at the wB97x-D/aug-cc-pVDZ level of theory for the  $S_1$  and  $T_1$  states of the two OH rotamers (given labels “a” and “b”) of both YGGFL A and FGGYL. We confirmed the locally excited nature of these states using natural transition orbital (NTO) representations for the  $S_0-S_1$  and  $S_0-T_1$  transitions. Note, first, that the calculations predict an electronic frequency shift ( $-338\text{ cm}^{-1}$ ) close to that observed between the  $S_0-S_1$  origin transition of FGGYL ( $-325\text{ cm}^{-1}$ ) relative to YGGFL A. This frequency shift is due to the  $\text{NH}_3^+\cdots\pi$  H-bond that is present in YGGFL A but absent in FGGYL. Second, the two OH rotamers of FGGYL are separated from one another by  $57\text{ cm}^{-1}$  in the calculation ( $-395$  vs  $-338\text{ cm}^{-1}$ ), confirming an assignment of the main and minor conformers of FGGYL to the two OH rotamers of FGGYL, which are separated by  $46\text{ cm}^{-1}$  in the UV photofragment spectrum. Finally, the calculations predict that the two OH rotamers of YGGFL A are separated by  $105\text{ cm}^{-1}$ . While we do not have a definite assignment, there is a weak transition about  $100\text{ cm}^{-1}$  to the red of the main electronic origin that may be due to this OH rotamer.

**Table 1. Experimental and Scaled Vertical Energy Differences for the  $S_0-S_1$  and  $S_0-T_1$  Transitions of the Two Tyr OH Rotamers (Labeled “a” and “b”) of YGGFL A and FGGYL**

| conformer           | $\Delta E(S_0-S_1)^a$<br>(cm) | electronic frequency shift<br>( $\text{cm}^{-1}$ ) <sup>b</sup> | $\Delta E(S_0-T_1)^a$<br>eV |
|---------------------|-------------------------------|-----------------------------------------------------------------|-----------------------------|
| YGGFL A expt        | 4.425 (35,684)                | 0                                                               | n/a                         |
| YGGFL a             | 4.425                         | 0                                                               | 3.484                       |
| YGGFL b             | 4.412                         | -105                                                            | 3.477                       |
| FGGYL expt<br>main  | 4.384 (35,359)                | -325                                                            | n/a                         |
| FGGYL expt<br>minor | 4.379 (35,314)                | -371                                                            | n/a                         |
| FGGYL a             | 4.383                         | -338                                                            | 3.478                       |
| FGGYL b             | 4.376                         | -395                                                            | 3.468                       |

<sup>a</sup>Scaled, vertical TDDFT calculations at the optimized ground state geometry, carried out at the wB97X-D/aug-cc-pVDZ level of theory, with scale factor 0.8593, which brings the calculated vertical energy differences into agreement with the known  $S_0-S_1$  origin of YGGFL A rotamer “a”. <sup>b</sup>The shift in the  $S_0-S_1$  origin of the indicated conformer relative to the  $S_0-S_1$  origin of YGGFL A rotamer “a”.

Based on the photofragment mass spectra taken following UV excitation and UV-IR double-resonance excitation, there are three photochemical reaction channels that compete with one another involving Tyr loss ( $-\text{Tyr}$ ), loss of neutral  $\text{NH}_3$  ( $-\text{NH}_3$ ), and peptide backbone fragmentation. Given the sizes of YGGFL and FGGYL and the multiconfiguration nature of the processes, particularly electron transfer, we focus attention in the calculations on establishing the energies of  $-\text{Tyr}$  and  $-\text{NH}_3$  asymptotes and on correlating the  $\pi\pi^*$   $S_1$  and  $T_1$  states with these asymptotes.

Table 2 lists the calculated energy asymptotes for Tyr loss relative to the optimized energy of the YGGFL A and FGGYL

**Table 2. Calculated Ground State Dissociation Asymptotes (eV) for the  $\text{NH}_3$  Loss and Tyr Side Chain Loss Product Channels in OH Rotamer “a” of Protonated YGGFL A and FGGYL<sup>a</sup>**

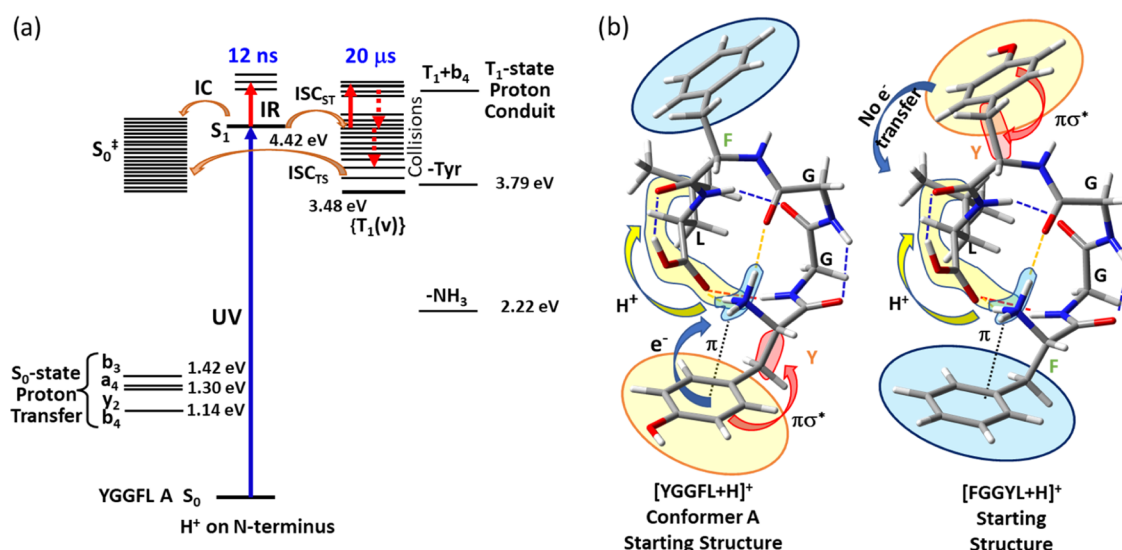
| conformer | $-\text{Tyr}$ (eV) | $-\text{NH}_3$ (eV) |
|-----------|--------------------|---------------------|
| YGGFL A   | 3.79               | 2.22                |
| FGGYL     | 3.44               | n/a                 |

<sup>a</sup>Fragment geometries optimized at the B3LYP-D3(BJ)/6-31+G(d) level of theory.

starting structures. Both product thresholds are well below the  $S_1$  origin in energy. However, in the Tyr loss channel, the  $S_1$  state correlates to an electronic excited state of the  $\text{CH}_2\text{-Ph-OH}$  radical about 2 eV above the ground state of the radical, while the lowest triplet state,  $T_1$ , correlates to the ground state radical. A similar analysis of the correlation diagram is not possible in the  $\text{NH}_3$  loss channel, which involves an electron transfer.

## 5. DISCUSSION

A primary strategy of this work is to use the backbone of a protonated peptide as a scaffold to initiate conformation-specific and site-specific photofragmentation from a well-defined, experimentally determined starting structure. Using a combination of UV alone and UV-IR double resonance, we have mapped out the spectroscopic signatures of the excited



**Figure 10.** (a) Schematic energy level diagram of photophysical and photochemical pathways of protonated YGGFL and FGYYL peptide ions following UV excitation (blue arrow) or combined UV/IR excitation (IR = red arrow) out of the  $S_1$  state or the  $T_1(v)$  state formed by intersystem crossing ( $ISC_{ST}$ ). IC = internal conversion,  $ISC_{TS}$  = reverse ISC from  $T_1$  to  $S_0$ . (b) Pictorial summary of the processes involved out of conformer A of YGGFL and the analogous chromophore-swapped structure of FGYYL.

state(s) involved and used those unique absorptions together with the time delay between UV and IR excitation pulses to determine the time scales and the relationships between site-specific initial excitation and the final products that are formed. **Figure 10a** presents a pictorial summary of the excited state processes involved in the site-specific and conformation-specific photofragmentation of protonated YGGFL(A) and FGGYL, with starting structures shown in **Figure 10b**.

One of the key results of the present study is the determination that the Tyr OH stretch fundamental in YGGFL(A) and FGGYL acts as a state-selective probe of the electronic character of the Tyr chromophore. In the  $S_0$  state, the OH stretch fundamental appears at  $3649\text{ cm}^{-1}$ , while the same transition in the  $S_1$  state (at the origin) shifts by  $73\text{ cm}^{-1}$  to the red to  $3576\text{ cm}^{-1}$ . The long-lived triplet state has its unique OH stretch at  $3615\text{ cm}^{-1}$ , roughly halfway between the  $S_0$  and  $S_1$ -state values. Using these unique OH stretch frequencies, we have mapped out the time-dependence of the  $S_1$  and  $T_n(v)$  OH stretch signals as a function of IR-UV delay (Figure 4b), providing a measure of the  $S_1$  lifetime for YGGFL(A) of  $12.1 \pm 2\text{ ns}$  and the triplet state rise of  $10.8 \pm 2\text{ ns}$ , equal to within error. The lifetime of this triplet state under our experimental conditions is  $\tau = 20 \pm 2\text{ }\mu\text{s}$ , a value that likely depends to some extent on the rate of collisional cooling inside the cold trap. A likely pathway out of this triplet state is via a second intersystem crossing ( $\text{ISC}_{T_S}$ ) to the singlet ground state.

In the Supporting Information (Section S11), we explicitly consider an alternative assignment for the long-lived OH stretch transition at  $3615\text{ cm}^{-1}$ ; namely, that it is due to a metastable ion-radical complex formed by recapture of the Tyr side chain,  $\dot{\text{C}}\text{H}_2\text{--Ph--OH}$ , by its attraction to the charged fragment that remains. This intriguing possibility, first put forward by Soorkia et al.<sup>2,32</sup> of a roaming-like recapture process is energetically plausible, as the calculated asymptote for radical loss is only about 0.6 eV below the  $\text{S}_1$  zero-point level. However, there are both spectroscopic and kinetic arguments against this alternative assignment, most notably in the fact that the  $\text{b}_4$  product also has this same OH stretch absorption

with identical lifetime characteristics to the corresponding band in the Tyr loss channel, pointing to these two product channels sharing a similar intermediate state. In what follows, we assume that the triplet state of the intact ion is responsible for the OH stretch transition at  $3615\text{ cm}^{-1}$ .

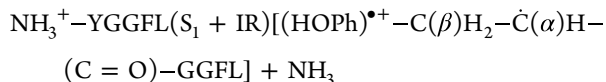
**5.1. Photochemistry at the  $S_1$  Origin.** Following electronic excitation to the  $S_1$  zero-point levels of YGGFL(A) and FGGYL, two photofragment products dominate the observed mass spectra (Figure 7); namely, (i) loss of the Tyr side chain at which electronic excitation originates and (ii) formation of the  $a_4/b_3/y_2$  peptide backbone fragment ions that are characteristic of hot ground state ions following internal conversion. Note that this UV-induced peptide backbone fragmentation occurs at several amide bonds along the peptide backbone, and is therefore rather indiscriminate, as is often desired in peptide sequencing.<sup>35</sup>

None of these backbone fragment ions are enhanced by IR excitation in UV-IR scans, consistent with the fact that they are produced by internal conversion, which produces ground state ions with 4.4 eV of vibrational energy. This energy is nearly 10 times that of a single IR photon, which could thereby mask the effect of IR excitation once internal conversion has occurred. Furthermore, once on the ground state surface, fast isomerization will shift and broaden IR absorptions. Finally, the peptide backbone fragments that are formed have thresholds for their formation of 1.1–1.4 eV (Figure 10a), far below the total energy in the ion, making it possible that fragmentation on the ground state surface may occur prior to delayed IR excitation.

The loss of the Tyr side chain ( $-\text{Tyr}$ ) is an excited state process since it is not observed in mass spectrometry studies that fragment YGGFL by collision-induced dissociation (CID).<sup>33</sup> This fragmentation is ubiquitous following UV excitation of Tyr-containing peptides,<sup>2</sup> and YGGFL(A) and FGYYL are no exception. As pointed out in previous work,<sup>32</sup> in order to break the  $\text{C}(\alpha)-\text{C}(\beta)$  bond, the  $\pi\pi^*$   $\text{S}_1$  or  $\text{T}_1(\text{v})$  states must couple to a  $\pi\sigma^*$  state associated with this bond. This side chain loss is also a dominant product channel following IR excitation under both  $\text{S}_1$  and  $\text{T}_n(\text{v})$  conditions.

## 5.2. IR-Enhanced Photochemistry in the Excited State(s). 5.2.1. Electron Transfer from Tyr\* to NH<sub>3</sub><sup>+</sup>.

One of the most striking results of the present work is the detection of the [MH<sup>+</sup>–NH<sub>3</sub>] product (*m/z* 539 Da) following loss of neutral NH<sub>3</sub> as a major photoproduct that is enhanced strongly by IR excitation in the S<sub>1</sub> state of YGGFL(A). Since the N-terminus was originally protonated (i.e., R–NH<sub>3</sub><sup>+</sup>), the loss of neutral NH<sub>3</sub> requires electron transfer from the S<sub>1</sub> excited state of the Tyr side chain to R–NH<sub>3</sub><sup>+</sup>, forming a 4-coordinate R–NH<sub>3</sub>, which subsequently dissociates by breaking the C(α)–N bond to form neutral NH<sub>3</sub> and leaving, at least initially, an ion in which the charge is on the phenol ring, with a radical site formed by breaking the C(α)–N bond:



In YGGFL(A), the Tyr  $\pi$  cloud and NH<sub>3</sub><sup>+</sup> group engage in a  $\pi$  H-bond, with a short distance for ET ( $\sim 2.7$  Å). In UV photodissociation studies of Tyr-containing peptides by Soorkia et al. that reached well above the S<sub>1</sub> origin, the analogous photochemical pathway is ascribed to a  $\pi\sigma^*(\text{NH}_3)$  charge transfer (CT) excited state involving promotion of a  $\pi$  electron on Tyr to a  $\sigma^*$  orbital on an N–H bond of NH<sub>3</sub><sup>+</sup> group.<sup>2</sup> Accessing this  $\pi\sigma^*(\text{NH}_3)$  state occurs most readily when one or more of the NH<sub>3</sub><sup>+</sup> hydrogens is not engaged in a H-bond. When no such vacancy exists, as is the case in protonated YGGFL with its two NH<sub>3</sub><sup>+</sup>...O=C H-bonds and one NH<sub>3</sub><sup>+</sup>... $\pi$  H-bond, the  $\pi\sigma^*$  state is destabilized. This suggests that, while the overall reaction is highly exoergic already at the S<sub>1</sub> origin (Figure 10a), the charge transfer (CT) state that leads to NH<sub>3</sub> loss is about 3500 cm<sup>−1</sup> above the S<sub>1</sub> origin in the Franck–Condon region of the  $\pi\pi^*$  states, thereby producing the enhancement in the mass-resolved UV-IR spectrum in the NH<sub>3</sub> loss channel at these energies. Below this energy, electron tunneling to the  $\pi\sigma^*(\text{NH}_3)$  may occur, enabling detection of the product with reduced efficiency.

Notably, in the gas phase ion, the preferred peptide backbone conformation is anticipated to change dramatically as the charge site shifts from NH<sub>3</sub><sup>+</sup> to Tyr<sup>+</sup> during electron transfer. In the starting structure (Figure 10b), the NH<sub>3</sub><sup>+</sup> group is in a strong binding pocket formed by H-bonds to two nearby C=O groups and the Tyr  $\pi$  cloud. The optimized structure following removal of neutral NH<sub>3</sub> from the known ground state structure of YGGFL A is shown in Figure S10, which we have used in computing the dissociation asymptote for NH<sub>3</sub> loss in Figure 10a. Following electron transfer, the polar groups in the peptide backbone reorganize to self-solvate the Tyr radical cation, a much more diffuse cation at the surface of the peptide structure.

One of the clearest indications of site-specific photochemistry in the present work is that NH<sub>3</sub> loss that is a dominant photoproduct in UV-IR spectra of protonated YGGFL(A) is not observed in protonated FGYYL either at the S<sub>1</sub> origin or following UV-IR excitation. Indeed, we have established that the small IR enhancement that is observed in FGYYL is at MH<sup>+</sup>–18 rather than MH<sup>+</sup>–17, and is due to H<sub>2</sub>O loss, not NH<sub>3</sub> loss. The quenching of the NH<sub>3</sub> loss product channel is a simple consequence of the decreased efficiency of electron transfer due to its steep falloff with electron donor–acceptor distance.<sup>36–38</sup> The Tyr electron donor in FGYYL is located on the opposite side of the 3D

structure (Figure 10b), with a 7.4 Å distance to the N-terminal nitrogen.

We found it initially puzzling that the S<sub>1</sub> state of YGGFL A shows strong IR-enhanced loss of NH<sub>3</sub>, while the corresponding UV-IR mass spectrum of the triplet state shows only very weak NH<sub>3</sub> loss (purple trace, Figure 8a). We have postulated that, in the S<sub>1</sub> state, IR excitation accesses vibrationally excited states near or above the crossing region with the  $\pi\sigma^*(\text{NH}_3)$  state that leads to NH<sub>3</sub> loss. However, the T<sub>1</sub> state accessed by ISC is  $\sim 0.9$  eV lower in energy than the S<sub>1</sub> state. This energy difference becomes vibrational energy in the triplet state. The vibrational density of states in the triplet manifold will be dominated by low-frequency modes that are not localized on the Tyr chromophore. As a result, we postulate that IR excitation out of T<sub>1</sub>(v) does not promote coupling to the  $\pi\sigma^*(\text{NH}_3)$  state in the way that the S<sub>1</sub> origin does, thereby shutting off the NH<sub>3</sub> loss channel in the T<sub>1</sub>(v) state.

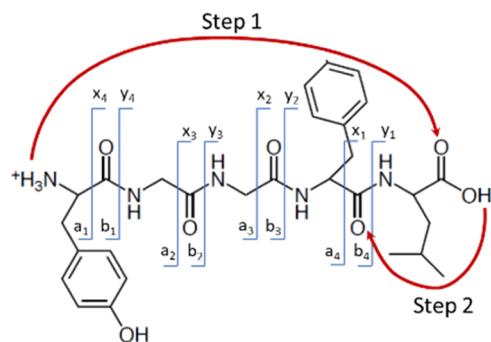
**5.2.2. Selective Backbone Cleavage: Proton Transfer along a Proton Conduit.** Peptide backbone fragmentation is a process of considerable practical importance in mass spectrometry, as it lies at the heart of sequencing of peptides and proteins.<sup>35</sup> As a result, the fragmentation of protonated peptides has been studied in detail in developing rules to understand the abundances of fragment ions.<sup>35,39,40</sup> The foundational insight on which much of this understanding lies is that the energized ion releases a mobile proton that can move along the peptide backbone to individual amide groups from which fragmentation occurs. This notion is encapsulated as “mobile proton theory”<sup>35</sup> which was subsequently refined into a “pathways in competition” framework by Paizs and Suhai,<sup>39</sup> which accounts for abundances of ions with charge localized on N-terminal (*a/b/c*) or C-terminal (*x/y/z*) fragments. Computational efforts to place the mobile proton model on a more rigorous basis, including how they are affected by the peptide’s 3D structure, are ongoing.<sup>41</sup>

In Section 4, we pointed out the selective formation of the b<sub>4</sub><sup>+</sup> and b<sub>4</sub><sup>+</sup>–NH<sub>3</sub> ions that involve breaking the amide bond that separates the Tyr and Leu residues. This result is highly unusual in two respects. First, bond breaking occurs at a specific amide bond to form b<sub>4</sub><sup>+</sup> exclusively, even though the initial energy of the reactant ion is much higher than the ground state dissociation asymptotes for all of a<sub>*m*</sub>, b<sub>*n*</sub>, and y<sub>*n*</sub> ions, as shown in the energy diagram of Figure 10a. These dissociation thresholds come from careful modeling of collision-induced dissociation (CID) and surface-induced dissociation (SID) measurements of protonated YGGFL by Laskin.<sup>33,42</sup> Second, while the b<sub>4</sub><sup>+</sup> ion is also formed in collision-induced (CID) and surface-induced dissociation (SID) of ground state [YGGFL + H]<sup>+</sup> ions, there is typically a strong correlation between the intensity of the a<sub>4</sub> and b<sub>4</sub> ions, with b<sub>4</sub> ions losing CO to form a<sub>4</sub> at high internal energies. Indeed, at the S<sub>1</sub> origins of YGGFL(A) and FGYYL, the a<sub>4</sub> ion dominates over b<sub>4</sub> (Figure 7a), so much so that the latter is barely discernible in the mass spectrum. Since the b<sub>4</sub> ion is susceptible to loss of CO to form a<sub>4</sub>, we surmise that some of the energy in the ion that leads to b<sub>4</sub> products is tied up in electronic excitation of the long-lived T<sub>*n*</sub>(v) states, limiting such fragmentation. Finally, the b<sub>4</sub> fragment ion is selectively formed following IR excitation in both protonated YGGFL and FGYYL, with the latter showing this selectivity more clearly (Figure 8b), since its formation does not compete with NH<sub>3</sub> loss, as it does in YGGFL.

We propose that the highly selective formation of  $b_4$  fragment ion is a direct consequence of the unique conditions present in this experiment. The starting structure is a single, well-defined H-bonded conformation, with the proton on the N-terminus. Due to the cryo-cooling, YGGFL and FGGYL are initially in their ground state zero-point levels of this single conformation. Excitation is accomplished through a combination of UV excitation to the  $S_1$  zero-point level of either Y(1) in YGGFL or Y(4) in FGGYL, followed by IR excitation of one of the XH bonds, placing a well-defined amount of vibrational energy ( $\sim 0.4$  eV) into the  $S_1$ -state ion. If intersystem crossing to the triplet manifold is the initial step in the formation of the  $b_4$  product, and if higher triplet states internally convert to  $T_1$  efficiently, then the ion will gain vibrational energy equal to the  $S_1$ – $T_1$  energy gap ( $\sim 0.9$  eV in the Franck–Condon region). When the energy of an IR photon is added, the vibrational energy available in the triplet state is about 1.3 eV, which is likely close in energy to the asymptote for fragmentation to form  $b_4$  with Tyr in its  $T_1$  state (see Figure 10a,  $T_1 + b_4$  asymptote).

As a result, we propose that this amount of energy in the  $T_1$  state is sufficient to launch the mobile proton from the N-terminus, but not indiscriminately. As the shading in yellow in Figure 10b highlights, there is a proton conduit that acts as a low-energy pathway for the proton to transfer from  $NH_3^+$  to the C=O group of amide(4) along the H-bonding network that is already present in the YGGFL and FGGYL starting structures. The H-bonds that make up this proton conduit are shown in Scheme 1 below. In step 1, proton transfer occurs

**Scheme 1. Proton Transfer Steps along Proton Conduit Leading to  $b_4$  Products**



from the  $NH_3^+$  group to the carbonyl oxygen on the C-terminal COOH group, to which it is H-bonded. As the C=O group accepts the proton, the OH proton can shuttle its proton to the amide(4) carbonyl O, producing the protonated C=O group that leads to fragmentation of the C–N bond in amide(4), forming the  $b_4$  cation.

We postulate, then, that the  $b_4^+$  fragment is the preferred and perhaps sole energetically allowed pathway for proton transfer from this well-defined starting geometry for a triplet state parent ion with 0.9 eV vibrational energy.

**5.2.3. Tyr Side Chain Loss.** Loss of the Tyr side chain is a unique excited state process that occurs efficiently in many Tyr-containing peptides following UV excitation.<sup>2</sup> Since its dissociation asymptote is more than 2 eV above the peptide backbone fragmentation thresholds, loss of the Tyr side chain is not observed in collision-induced or surface-induced dissociation. We have seen that both YGGFL(A) and FGGYL lose the Tyr side chain as a major photoproduct,

despite occurring out of very different site-specific environments, with Y(1) in YGGFL(A) in close interaction with the  $NH_3^+$  site and Y(4) in FGGYL nearly free. The potential energy curves for dissociation of the C( $\alpha$ )–C( $\beta$ ) bond in Figure 10a, while not of quantitative accuracy, nevertheless show a mechanistic pathway by which formation of the triplet state can lead efficiently to Tyr side chain loss at its ground state asymptotic limit. Near the Franck–Condon region, the  $S_1$  and  $T_1$  states are separated by 0.9 eV, but a small elongation of the C( $\alpha$ )–C( $\beta$ ) bond brings  $S_1$  and  $T_1$  into near degeneracy. Further elongation of the C( $\alpha$ )–C( $\beta$ ) bond stabilizes the  $T_1$  state so that it becomes nearly isoenergetic with the  $S_0$  state, leading to dissociation at the  $S_0$ -state dissociation limit. Multiconfiguration calculations are needed to map out these surfaces more fully, a challenging task in a molecule of this size and complexity.

## 6. CONCLUSIONS

The present work adds to a small but growing list of photochemical studies of large, conformationally complex molecules<sup>2,10–14</sup> that provide an unusual level of control to the excitation process out of which photochemistry is initiated. In photochemical studies of comparable-sized molecules in solution and/or at room temperature in the gas phase, photoexcitation occurs out of a conformationally and energetically diverse ensemble. This leaves open the possibility that there may be subgroups of molecules in which the photochemical pathways could be dictated by the specific starting conformation and perhaps controlled to some extent by the precise electronic/vibrational energy content given to this starting conformation.

We have chosen to study protonated YGGFL and FGGYL peptide ions because they form a model pair to explore site-specific excitation out of a well-defined and experimentally determined starting structure. By cooling these gas phase ions in a 5 K cold trap, we remove any excess internal energy and trap the ions in the zero-point levels of their starting conformations. The cooling then enables excitation of this single conformation via its site-specific UV excitation wavelength. Once there, photofragmentation is explored simply by mass-resolving the photofragments. Furthermore, UV-IR double resonance is used to record the IR spectra in the excited singlet ( $S_1$ ) and triplet states ( $T_n(v)$ ), with the Tyr OH stretch serving as diagnostic of the electronic state. With this spectroscopic signature, we follow the initial dynamics of the  $S_1$  and  $T_1(v)$  states in real time.

We have observed the site-specific enhancement of electron transfer leading to loss of neutral  $NH_3$ , which occurs readily following IR excitation in YGGFL, in which the Tyr( $\pi\pi^*$ ) state is involved directly in a  $\pi$  H-bond with the  $NH_3^+$  group that serves as acceptor for the electron transfer. In FGGYL, where the Tyr is across the peptide scaffold from the  $NH_3^+$  group,  $NH_3$  loss is not observed.

One of the dominant fragmentation channels involves breaking the C( $\alpha$ )–C( $\beta$ ) bond of the Tyr residue, leading to loss of the Tyr side chain. The present study identifies the  $T_1$ -state as a probable intermediate state, but has not obtained spectroscopic evidence for the  $\pi\sigma^*$  state that leads to C( $\alpha$ )–C( $\beta$ ) bond breakage.

Finally, even the peptide backbone fragmentation channels, which are notoriously promiscuous in most circumstances, favor a single specific fragmentation site in YGGFL and especially in FGGYL following UV-IR double resonance,

leading to selective formation of the  $b_4^+$  fragment. We postulate that this selectivity arises from the unique starting structure, which contains a proton conduit between the  $\text{NH}_3^+$  “launch” site and amide(4) acceptor site, combined with the unique combination of electronic and vibrational excitation that limits the vibrational energy content so that the proton prefers this conduit for proton shuttling leading to specific backbone cleavage. This postulate requires further theoretical work to confirm, deny, or refine, but the experimental result constitutes an example of laser-induced, bond-specific dissociation, here appearing in a large, flexible peptide ion.

## ■ ASSOCIATED CONTENT

### SI Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/jacs.4c01576>.

Schematic diagram for UV-IR gain spectroscopy; conformation-specific IR-UV dip spectra of OH rotamers of FGGYL; ground state IR-UV spectrum of YGGFL conformer B; comparison of IR-UV gain and dip spectra for YGGFL; lifetime scans of FGGYL  $S_1$  and  $T_n(v)$  states; triplet state lifetime determination; peptide backbone fragmentation labeling scheme; high-resolution mass spectra demonstrating that YGGFL loses  $\text{NH}_3$  and FGGYL loses  $\text{H}_2\text{O}$ ; mass-resolved UV-IR spectra of FGGYL under  $S_1$ -state conditions; optimized structure of the biradical cation formed by loss of neutral  $\text{NH}_3$  from YGGFL A; and summary of evidence for the long-lived state as a triplet state (PDF)

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## Notes

The authors declare no competing financial interest.

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